NOVA SCHOOL OF
BUSINESS & ECONOMICS

Hedge Funds

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Assignment 1:
Portfolio Backtesting

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To construct the equity portion of the trading strategy we selected the following indices: S&P 500, Dow Jones, and Nasdaq 100. These equity indexes figure among the most liquid futures contracts globally. The fixed income component utilizes US Treasury bond futures across three key maturities (2-y, 5-y, and 10-y), providing broad exposure along the yield curve. As the world's deepest sovereign bond market, this ensures high liquidity and allows the strategy to capture not just duration beta but also curve steepening and flattening dynamics. Furthermore, combining these bond and equity instruments enables us to analyse the U.S. market in a comprehensive manner while reducing reliance on any single asset, thereby improving diversification. The commodities chosen, Gold, Silver, and Copper, represent a strategic mix of precious and industrial metals. On one hand, Gold serves as a classic safe-haven asset, while Copper can act as a cyclical proxy for global economic growth due to its use in industrial machinery and electrical equipment.

For each asset class, both Trend Following (TF) and Mean Reversion (MR) strategies were tested using varying moving average windows and volatility bands. Within each asset class, individual futures returns were equally weighted to reduce idiosyncratic noise and considering that the equal weight portfolio is an extreme shrinkage estimator of the optimal portfolio which is much simpler to implement [1]. The results (*Table 1*) indicated distinct characteristics: Equities performed best under a MR framework (20 day, 2.0σ), suggesting that over shorter, monthly horizons, equity markets are prone to behavioural overreaction and liquidity-driven noise, components that are mean-reverting by construction. Conversely, Bonds and Commodities exhibited stronger persistence, favouring TF strategies with longer (200 day, 1.5σ) and medium (50 day, 1.5σ) windows, respectively, given their influence from macroeconomic cycles and rewarding sustained momentum. These optimized asset class returns were aggregated into a portfolio using two distinct allocation strategies: Capped Weight Tangency and a Risk Parity (RP) (*Table 2)*. The Tangency portfolio was constructed via Mean-Variance Optimization (MVO) to maximize the historical Sharpe ratio subject to structural concentration limits. A 50% cap was applied to both Equities and Bonds to prevent a single asset class from dominating the portfolio's risk profile, while Commodities were restricted to 25% to mitigate the impact of idiosyncratic volatility and frequent supply-side shock. This constrained optimization resulted in a performance-oriented allocation that saturated the bond limit (50%) and maintained significant equity exposure (36.5%). In contrast, the RP approach produced a highly defensive allocation with approximately 68% in bonds, 20% in equities, and only 12% in commodities.

To identify patterns of superior performance, the strategies were stress-tested under three distinct macroeconomic regimes: Market Trend, Interest Rate Shifts, and Volatility (*Table 3)*. Analysis of the Market Trend (proxied by the S&P 500 relative to its 200-day moving average) confirmed that the Tangency portfolio significantly outperforms during Bull Markets, leveraging its higher equity beta to capture upside momentum. Conversely, in Bear Markets, the defensive structure of Risk Parity offered superior capital preservation. The strategies were then tested against Interest Rate regimes (using 10-Year Treasury Yield changes). While Risk Parity typically struggles in rising-rate environments due to high duration, the trend-following nature of the underlying bond strategies helped mitigate losses compared to passive holdings. Finally, regarding Volatility (proxied by the VIX), the Tangency portfolio consistently had higher returns than the RP allocation, for VIX thresholds above 15. With a bigger exposure to volatile asset classes (Equities, Commodities), the tangency allocation is able to capitalize on market overcorrections with higher raw returns Particularly, during periods of extreme market stress (VIX > 50), the equities MR aggressive 2.0 σ trigger is activated more often. However, the tangency allocation shows larger drawdowns in extreme downturns, such as 2008 and 2011, thus materially weakening the risk-adjusted returns, Info Sharpe, and increasing tail risk.

Building upon this asymmetric risk profile, a dynamic Regime-Switching strategy was implemented to capitalize on the specific strengths of each allocation method. This adaptive approach toggles between the two static portfolios based on a volatility threshold: deploying the aggressive Tangency weights when the VIX trades below 25 to maximize growth during stable trends and switching to the defensive Risk Parity weights when the VIX rises above 25 to prioritize capital preservation. The backtesting results (*Table 4*) validated the superiority of this dynamic framework. The Regime-Switching strategy achieved an annualized Information Sharpe Ratio of 0.690, decisively outperforming both the static Risk Parity portfolio (0.626) and the Tangency portfolio (0.611). Crucially, the maximum drawdown was contained at -4.49%, an improvement over the -5.72% suffered by the static Tangency portfolio. Finally, sensitivity analysis (*Table 5*) confirms the model's robustness across a 20-30 volatility range, suggesting that while the standard 25 threshold is effective, a slightly more sensitive defensive trigger could yield even higher risk-adjusted efficiency.

Appendix

Table 1

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STEP 1 & 2: OPTIMIZING STRATEGIES PER ASSET CLASS
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Equities      | MR | MA= 20 σ=2.0 | Sharpe= 0.427
Bonds         | TF | MA=200 σ=1.5 | Sharpe= 0.345
Commodities   | TF | MA= 50 σ=1.5 | Sharpe= 0.270

Asset Class Strategy  Window  Std Dev   Sharpe  Tickers
   Equities       MR      20      2.0 0.427093        3
      Bonds       TF     200      1.5 0.344873        3
Commodities       TF      50      1.5 0.270327        3

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Table 2

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STEP 3: PORTFOLIO OPTIMIZATION
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Optimized Weights:
            Tangency Risk Parity
Equities       36.5%       20.1%
Bonds          50.0%       68.0%
Commodities    13.5%       12.0%

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Table 3

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STEP 4: IDENTIFYING REGIME INDICATORS
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--- Analysis: Market Trend (S&P 500) ---
              Ann. Spread Daily Volatility  Days Observed Win Rate
Regime_Growth
Bear Market         0.59%            2.13%           1305   35.40%
Bull Market         0.36%            0.73%           3507   40.52%
> Insight: Tangency performs best in 'Bear Market' regime.

--- Analysis: Interest Rate Environment (10Y Yield) ---
              Ann. Spread Daily Volatility  Days Observed Win Rate
Regime_Rates
Falling Rates       0.70%            1.62%           2498   38.71%
Rising Rates        0.12%            0.74%           2314   39.59%
> Insight: Tangency performs best in 'Falling Rates' regime.

--- Analysis: Volatility Levels (VIX) ---
               Ann. Spread Daily Volatility  Days Observed Win Rate
Regime_Vol
Low (<15)           -0.11%            0.38%           1546   37.06%
Medium (15-25)       0.57%            0.77%           2400   40.96%
High (25-50)         0.85%            2.25%            791   38.18%
Extreme (>50)        1.88%            5.39%             75   33.33%
> Insight: Tangency performs best in 'Extreme (>50)' regime.

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Table 4

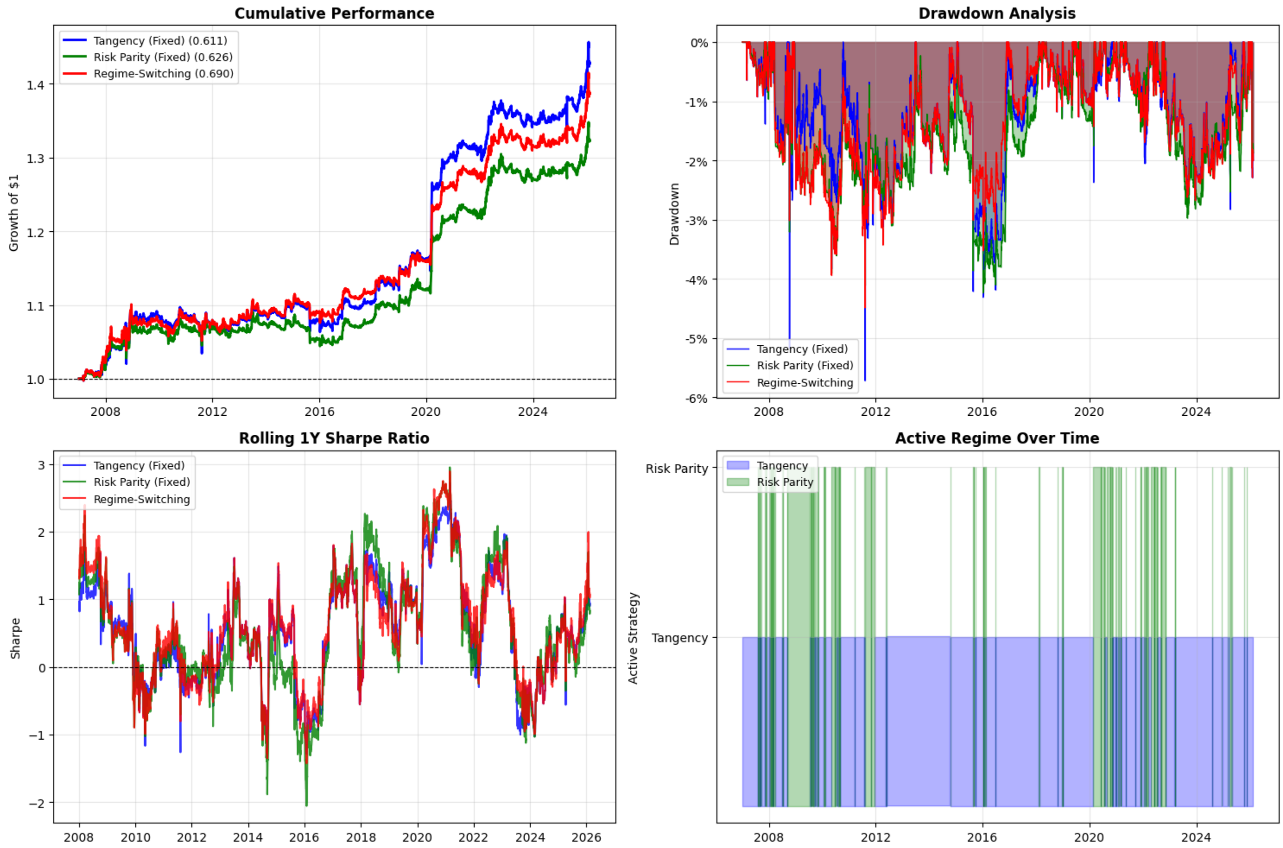
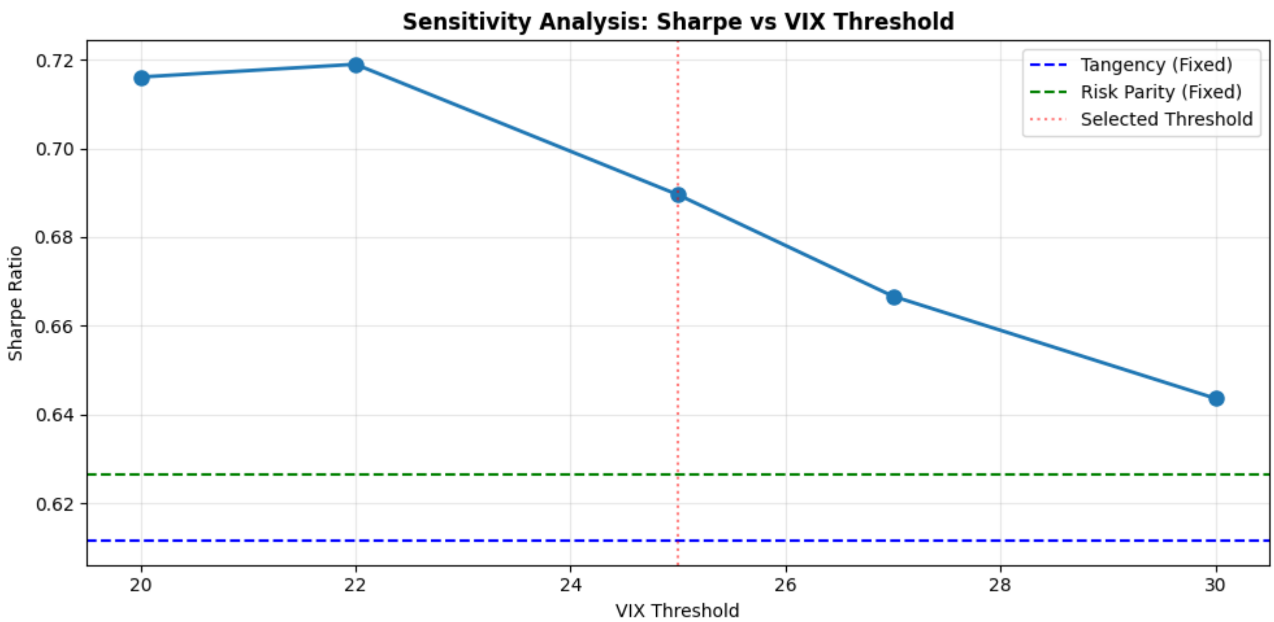


Table 5



References

[1] DeMiguel, V., Garlappi, L., & Uppal, R. (2009). Optimal Versus Naive Diversification: How Inefficient is the 1/N Portfolio Strategy? The Review of Financial Studies, 22(5), 1915–1953. https://doi.org/10.1093/rfs/hhm075